

## **CariSurg MedTech Pathways Research Project Preliminary Proposal Project Title**

AI-Assisted Risk Flagging for Emergency Department Triage in Resource-Constrained Settings

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### **Statement of the Problem**

Emergency departments often deal with overcrowding, time pressure, and variation in triage decisions, especially when nurses must quickly identify patients who may be at higher risk (Da'Costa et al., 2025; De Freitas et al., 2020; Tyler et al., 2024). Although recent studies show that AI can support emergency triage, many are based on retrospective or single-site datasets and do not always show how the output would fit into real bedside decision-making (Agius et al., 2025; Tahernejad et al., 2024; Wong et al., 2021). The key problem is that it remains unclear how a simple, nurse-facing AI risk flag could be validated and integrated into triage workflows in a resource-constrained emergency department without increasing alert burden or replacing clinical judgement.

### **Previous Work**

Tyler et al. (2024) reviewed how artificial intelligence has been used in hospital emergency department triage. Their review showed that AI tools may help with identifying high-risk patients earlier, improving prioritisation, and supporting resource allocation. However, one important point from the review is that many studies focus mainly on model performance, while less attention is given to how these tools would actually work in a busy clinical environment. This is relevant to the proposed pilot because a tool may perform well in testing but still needs to be practical and safe for triage nurses to use (Tyler et al., 2024).

Da'Costa et al. (2025) discussed the potential of AI-driven triage in emergency departments. The review highlighted possible benefits such as faster patient prioritisation, reduced waiting times, and better use of limited resources. At the same time, the authors raised concerns about data quality, bias, clinician trust, ethics, and implementation. These concerns are important because an AI triage tool should support the nurse's judgement rather than replace it (Da'Costa et al., 2025).

Araouchi and Adda (2024) presented TriageIntelli, an AI-assisted multimodal triage system for health centres. Their study compared several machine learning models, including random forests, support vector machines, artificial neural networks, gradient boosting, logistic regression, XGBoost, and a stacking model. The results suggested that machine learning can predict triage categories with promising accuracy. However, this also shows why accuracy alone

is not enough, since a useful triage tool must also be understandable, usable, and tested in the setting where it will be applied (Araouchi & Adda, 2024).

Agius et al. (2025) used emergency department data from Mater Dei Hospital in Malta to predict Emergency Severity Index level, hospital admission, and admitting ward. Their study used XGBoost-based models and showed that routinely collected emergency department data can support both triage and resource planning. The findings also suggested that prediction performance can improve when additional clinical information is included, rather than relying only on basic demographic or presenting information. This paper is useful for the proposed pilot because it shows how ED data can support decision-making, but its findings would still need to be validated before being used in a different emergency department context (Agius et al., 2025).

Tahernejad et al. (2024) reviewed the use of artificial intelligence in triage during emergencies and disasters. The review examined studies involving AI-supported triage, machine learning, smart triage technologies, and emergency decision support. The paper showed that AI may help with response time, resource management, real-time data transfer, and vital sign monitoring in high-pressure emergency settings. However, it also identified practical barriers such as staff training, trust in AI systems, equipment limitations, privacy concerns, and implementation challenges. This connects well with the Mercer scenario because a resource-constrained emergency department would need a tool that is simple, realistic, and designed around actual workflow limitations (Tahernejad et al., 2024).

De Freitas et al. (2020) explored patient flow challenges in a Caribbean emergency department. Their study showed that emergency care can be affected by overcrowding, delays, limited resources, and process bottlenecks across the patient journey. This is important for the proposed pilot because an AI-assisted triage tool should not only be technically accurate, but also realistic for the workflow and resource pressures of a Caribbean emergency department context (De Freitas et al., 2020).

Wong et al. (2021) externally validated a widely implemented sepsis prediction model and found that real-world performance was weaker than expected. The study highlights why clinical prediction tools should not be accepted based only on internal performance claims, especially when poor calibration or alert burden may affect clinical decision-making. This supports the need for local validation, human oversight, and careful monitoring before using AI-assisted risk alerts in emergency triage (Wong et al., 2021).

## **Gaps Identified**

### **Gap 1: Limited local validation in resource-constrained emergency departments**

Many AI triage studies report promising results, but the models are often developed using retrospective or single-hospital data. This makes it difficult to know whether the same approach would work well in another hospital, especially in a resource-constrained Caribbean emergency department where patient flow, staffing, and system pressures may differ from larger international hospitals. A 12-week Mercer pilot can begin to address this by testing a simple risk-flagging approach using approved de-identified, simulated, or Mercer-style triage data before considering wider use.

### **Gap 2: Limited evidence on workflow integration and clinical usefulness**

A second gap is the difference between model performance and real clinical usefulness. Several studies report accuracy, AUC, or model comparisons, but they do not always explain how a nurse would use the output during triage. In practice, the tool must be clear, quick to interpret, and safe to use. This pilot can address that by designing the output as a simple “urgent review” risk flag, along with the abnormal findings that triggered the flag, rather than trying to replace the triage nurse or automatically assign a final triage category.

## **Proposed Solution**

The proposed solution is a 12-week pilot of a simple AI-assisted risk-flagging tool for emergency department triage. The tool would use information that is normally available during triage, such as vital signs, age, presenting complaint, and basic clinical observations. Its purpose would be to highlight patients who may need urgent review, not to replace the triage nurse or make final clinical decisions.

The pilot would use de-identified historical triage data if approved and available. If approved local data are not available within the 12-week timeframe, the prototype could first be tested using simulated or Mercer-style triage data based on realistic emergency department variables. This would allow the project to test the workflow, safety rules, missing-data handling, and risk-flagging logic without making live clinical decisions.

The pilot would begin by deciding which input variables and safety rules should be included. The data would then be cleaned and checked for missing, unrealistic, or inconsistent values. A simple baseline model could be developed and compared with basic rule-based clinical thresholds. The output would be kept easy to understand by showing the patient’s risk flag and the main abnormal findings that contributed to it.

The success of the pilot would be judged using practical measures such as how well the tool identifies high-risk patients, how many false alerts it creates, how it handles missing data, and whether triage staff can understand the output. Within the 12 weeks, the expected deliverables would be a small prototype, a short evaluation report, and a recommendation on whether the idea should move forward for further testing. This keeps the project realistic, clinically relevant, and suitable for the timeframe.

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